PRODUCTION BASELINE v1.0

Plugin Lifecycle, Installation & Dependency Management

MASS-APP-015 / Volume 02

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Installation is governed presence, not execution authority. V02 binds every material lifecycle change to a validated V01 manifest, deterministic dependency graph, explicit human approval, tenant scope, and durable evidence.

1. Mission and Ownership

V02 converts a V01-validated plugin package into a controlled tenant installation. It owns lifecycle requests, prerequisite findings, dependency resolution, immutable plans, independent decisions, scoped installation records, health gates, rollback evidence, recovery, and safe-removal tombstones.

V02 OWNS	V02 CONSUMES	V02 NEVER CLAIMS
Plans, decisions, installation state, lifecycle evidence	V01 manifests and declarations; ENG identity, authorization, events, workflow, audit, configuration, API, deployment, entitlement and lineage	Runtime invocation (V03), deployment-engine execution, marketplace storefront, other applications' data

Permanent Architecture vs V1

Permanent architecture separates declared package truth, target-environment truth, deterministic resolution, human authority, execution coordination, and immutable evidence. V1 supports organization, workspace, project, and user scopes; required/optional/peer/conflict edges; install, activate, suspend, upgrade, downgrade, rollback, repair, and removal.

Human Authority

- A requester may request but cannot approve its own plan.
- Approval binds manifest digest, exact versions, target scope, permissions, migrations, steps, health gates, and rollback target.
- A material change creates a new plan revision and invalidates prior approval.
- A plugin, publisher, automation, or execution principal cannot self-approve.

Role Mapping

ROLE	BASELINE	AUTHORITY
Installation Viewer	Viewer	Read authorized state and evidence
Installation Requester	Contributor	Request scoped lifecycle action
Plugin Operator	Contributor extension	Execute an approved plan
Lifecycle Steward	Steward	Resolve dependencies and recommend recovery
Installation / Security Approver	Administrator	Independently decide bound plans
Executive Approver	Executive	Decide explicitly exceptional risk

2. Controlled Installation Flow

#	STAGE	CONTROL
01	REQUEST	Exact V01 version, digest, scope, grants, configuration refs, idempotency key
02	VERIFY	Eligibility, platform/application/environment prerequisites, entitlement, security
03	RESOLVE	Declared graph only; deterministic versions, conflicts, peers, optional omissions
04	PLAN	Immutable ordered steps, approvals, migrations, health gates, compensation, rollback
05	APPROVE	Independent human decision bound to the complete plan digest
06	EXECUTE	Authorized idempotent steps through owning platform gateways
07	ACTIVATE	Explicit action after grants, entitlement, migrations and readiness pass

Dependency Resolution

KIND	RULE	FAIL-CLOSED CONDITION
Required	Exactly one compatible published version	Missing, cycle, scope violation, entitlement failure
Optional	Only when explicitly selected	Never silently installed; omission impact recorded
Peer	Already present or explicitly included	Absent, incompatible version, or invalid scope
Conflict	No matching installed or planned version	Matching conflict remains unresolved

Deterministic Selection

- Retain a healthy compatible installed version; otherwise select the highest published, eligible, supported version satisfying every inbound range.
- Required edges resolve first; peer and conflict verification follows. Canonical plugin name and version are stable ordering tie-breakers.
- Transitive nodes retain provenance and reason chains. Required cycles block; optional-only cycles are omitted with a finding.
- Resolver output is advisory until the complete immutable graph is independently approved.

Scope Rule

A dependency must have equal or broader scope than its dependent and tenant policy must permit inheritance. Promotion or scope widening always creates a new plan and approval.

3. Lifecycle State and Recovery

Every transition requires expected state, installation revision, approved plan digest, actor authorization, and an idempotency key. A repeated command returns its original outcome. A stale revision fails without side effects.

PHASE	STATES	KEY GATE
Plan	requested / planning / awaiting_approval / approved	Validated digest, resolved graph, independent decision
Install	installing / installed	Durable step evidence and required readiness
Operate	activating / active / suspending / suspended	Explicit action; runtime remains V03
Change	upgrading / rolling_back / repairing	Approved target, checkpoints, health
Exit	removing / removed	Reverse dependency analysis and retained tombstone
Failure	recovery_required / failed	No false success; resume from verified checkpoint

Upgrade, Downgrade, Rollback, Repair

- Each change compares source/target digests, permissions, dependencies, configuration schema, data migrations, compatibility, health, and rollback feasibility.
- Permission expansion always requires new approval. Unsafe downgrade is rejected.
- Rollback uses an approved target and captured checkpoint; reverse migrations are never invented.
- Repair revalidates digest, dependencies, configuration refs, grants, and health, and replays only explicitly idempotent steps.

Partial Failure

Failure atomically records the completed-step boundary and durable audit/outbox evidence before entering recovery. Irreversible work is never hidden. If durable audit or outbox persistence fails, the state transition fails and remains retryable.

Safe Removal

Required dependents block removal. Optional dependents create impact findings. Sole-use dependencies become orphan candidates, never automatic deletions. V02 removes only its metadata/configuration references and retains a tombstone; application- or engine-owned data remains with its owner.

4. Platform Gateways and Integrity

GATEWAY	CONSUMPTION	FAILURE BEHAVIOR
Plugin Registry	V01 manifest, version, digest, declarations	Reject missing, unpublished or changed version
Identity / Authorization	Actor, role, scope and separation of duty	Fail closed without grant expansion
Workflow	Approval and execution coordination	Pause and retain completed-step evidence
Event / Audit	Outbox events and immutable evidence	Retry; lifecycle decision fails without durability
Configuration	Approved settings and secret references	Block readiness; never copy secret values
Deployment	Owned environment execution	Record failure; V02 never impersonates executor
Entitlement	License and entitlement evidence	Block activation when required evidence is invalid
Lineage	Package, plan, migration, rollback provenance	Block irreversible transition when mandatory
Administration / Security	Published organization/security contracts	Never fabricate unavailable contracts

Database Enforcement

- Seventeen tenant-owned tables use UUID defaults, UNIQUE(id, tenant_id), composite tenant-safe foreign keys, and auth.jwt()-derived RLS.
- Triggers prohibit requester self-approval, approval digest mismatch, destructive evidence mutation, and mutation of decided plans and terminal operations.
- Approval, migration checkpoint, rollback, removal, health, audit, and published event evidence is immutable at its governing state.
- Outbox payloads cannot be rewritten or deleted; publication time is the only allowed publication mutation before it becomes fixed.

Migration Ownership

V02 owns installation metadata and configuration-schema coordination. Source applications and engines own their data. V02 stores only owner, migration reference, checksum, preconditions, checkpoint, result, and compensation reference. Irreversible migrations require explicit warning, approval, and recovery evidence.

5. Contract Surface and Constitutional Boundary

ACTION	ENDPOINT	MINIMUM ROLE
Request	POST /plugin-installation-requests	Installation Requester
Resolve	POST /plugin-installation-requests/{id}/resolve	Lifecycle Steward
Approve / Reject	POST /plugin-installation-plans/{id}/{decision}	Installation Approver
Install / Activate / Suspend	POST /plugin-installations/{id}/{action}	Plugin Operator
Upgrade / Downgrade / Rollback / Repair	POST /plugin-installations/{id}/{action}	Plugin Operator
Remove	POST /plugin-installations/{id}/remove	Plugin Operator
Inspect	GET /plugin-installations/{id}/{dependencies|health|history}	Installation Viewer

Mutation Contract

Mutation endpoints require Idempotency-Key, expected revision where applicable, actor authorization, and correlation identifiers. Installation and rollback examples accompany this volume as machine-readable JSON. Events are written in the same transaction as authoritative state changes.

> **CONSTITUTIONAL BOUNDARY**
>
> V02 may validate, plan, compare, recommend, coordinate an authorized lifecycle action, and preserve evidence. It cannot expand privilege, install undeclared dependencies, self-approve, imply runtime execution authority, duplicate deployment ownership, bypass license/security/privacy controls, remove another owner's data, suppress failures, or make an irreversible change without the approved bound plan.

Artifact References

ARTIFACT	PURPOSE
MASS-APP-015-V02_Migration_Reference.sql	Tenant-safe schema, RLS, triggers, immutable evidence
MASS-APP-015-V02_Plugin_Lifecycle_State_Machine.mmd	Complete lifecycle states and transition paths
MASS-APP-015-V02_Dependency_Resolution_Rules.md	Deterministic edge, cycle, scope, and selection rules
MASS-APP-015-V02_Installation_and_Rollback_Contract_Examples.json	Machine-readable request, plan, rollback, and result examples
MASS-APP-015-V02_API_Inventory.csv / Data_Model.csv	Contract and persistence inventories